Daksh Upadhyay's project report cmpsc 445

Data Collection

1. Use selenium and Glassdoor + us gov to screen for jobs
2. We collect 3 attributes: [salary, company name, location]
3. Samples more than >30 at the time of running the scraper model, also getting blocked from cloudfare with Glassdoor we can use proxy

Data Preprocessing

1. The salary is conveted to int before its put into the dataframe
2. We use label encoder to encode the location and company name before feeding into the model

Training and inference

We train two models

1. **Logisitc regression**
2. **Linear regression**

```python
def train_linear_regression(X, y, title):
    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

    model = LinearRegression()
    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)
    mse = mean_squared_error(y_test, y_pred)
    print('mse: ', mse)
    # plot the results
    plt.scatter(y_test, y_pred)
    plt.xlabel('True Values')
    plt.ylabel('Predictions')
    plt.title('Linear regression ' + title)
    plt.show()

# train logisiic regression
def train_logistic_regression(X, y, title):
    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

    model = LogisticRegression()
    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)
    mse = mean_squared_error(y_test, y_pred)
    print('mse: ', mse)
    # plot the results
    plt.scatter(y_test, y_pred)
    plt.xlabel('True Values')
    plt.ylabel('Predictions')
    plt.title('Logistic regression ' + title)
    plt.show()
```

Visualization

We plot the true values against the predictions to see how far is the model from the actual values we also print the mean squared error for the model

Difficulties

I had the most difficulties with the web scraper and it wasn't working the first time then it started to work with Selenium and using Chrome as the driver. Some data still doesn't get loaded from Glassdoor because of cloudfare IP restrictions so we need to use proxy server